

Business Process Automation Platform

Enterprise-grade business process automation platform built on Java & Spring Boot.

Business Process Automation Platform lets teams design, run, and monitor multi-step business workflows — approvals, onboarding, procurement, claims processing — without hand-rolling state machines for every use case. It combines a BPMN-compliant workflow engine, a pluggable rules engine, and first-class integrations into a single self-hostable platform.

 Status: Early development — core workflow engine in progress.

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Why This Platform

Most teams end up scattering business logic across cron jobs, if/else chains, and ad-hoc status columns. This platform centralizes that logic into versioned, auditable process definitions — so a “process” (an approval chain, a claims pipeline, an employee onboarding flow) is a first-class object you can design, test, deploy, and monitor like code.

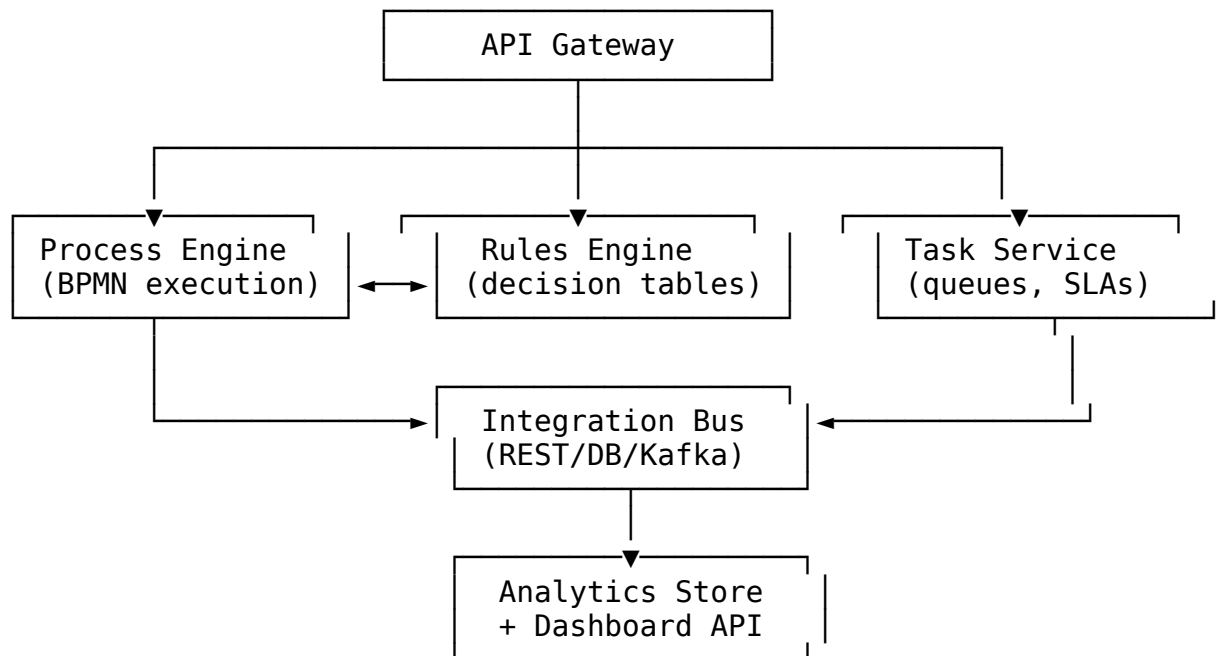
Key Features

- **Workflow / BPMN Engine** — Define processes as BPMN 2.0 diagrams or fluent Java DSL. Supports sequential, parallel, and conditional branching, sub-processes, timers, and compensation/rollback steps.
- **Task & Approval Management** — Human task queues with delegation, escalation, SLA timers, and multi-level approval chains (single approver, quorum, sequential sign-off).
- **Rules Engine** — Externalized decision logic (DMN-style tables or embeddable rule scripts) so business users can tune conditions without redeploying code.

- **Integrations** — Native connectors for REST/SOAP APIs, relational databases, and message brokers (Kafka/RabbitMQ), plus a webhook framework for outbound events.
- **Dashboard & Analytics** — Real-time process visibility: in-flight instances, bottleneck detection, average cycle time per stage, SLA breach alerts.
- **Multi-Tenant / Enterprise Ready** — Tenant isolation at the data and process-definition level, RBAC, audit logging, and horizontal scalability for high-throughput deployments.

Note: these are the platform's target capabilities. See Project Structure below for what's actually implemented today.

Architecture



This is the target architecture as the platform grows into separate modules. The current codebase is a single-module starter (see Project Structure).

Tech Stack

Layer	Technology
Language / Runtime	Java 21, Spring Boot 3.x
Workflow Engine	In-memory starter today; Flowable/Camunda planned
Persistence	Currently in-memory; PostgreSQL planned
Messaging	Kafka/RabbitMQ planned
Rules	Drools or DMN decision tables planned
Auth	Spring Security + OAuth2/JWT

Layer	Technology
	planned
API	REST (OpenAPI 3.0 spec)
Build	Maven
Containerization	Docker (planned)

Getting Started

Prerequisites

- Java 21+
- Maven 3.9+

Run locally

```
git clone https://github.com/Babinaroot/BUSINESS-PROCESS-AUTOMATION-PLATFORM.git
cd BUSINESS-PROCESS-AUTOMATION-PLATFORM
```

```
mvn spring-boot:run
```

The API will be available at `http://localhost:8080`.

Project Structure

Currently a single-module starter — will split into separate modules (engine, rules, tasks, integrations, analytics) as features grow. See Roadmap.

```
BUSINESS-PROCESS-AUTOMATION-PLATFORM/
├── src/main/java/com/flowforge/
│   ├── model/           # Process, step, and instance data models
│   ├── service/         # Core workflow logic (in-memory for now)
│   └── controller/      # REST API endpoints
├── src/main/resources/
│   └── application.yml
├── pom.xml
├── .gitignore
└── README.md
```

Configuration

Currently runs with no external dependencies — in-memory storage, no database or auth yet.

```
server:
  port: 8080
```

Database, messaging, and auth config will be added here as those features are implemented (see Roadmap).

API Overview

Endpoint	Description
POST /api/v1/processes	Create a new process definition
GET /api/v1/processes/{id}	Get a process definition
POST /api/v1/processes/{id}/start	Start a new process instance
GET /api/v1/instances/{id}	Get instance status and current step
POST /api/v1/instances/{id}/complete-current	Complete the current step, advancing the instance

Roadmap

- ☐ Persist definitions/instances to PostgreSQL
- ☐ Split into multi-module architecture (engine, rules, tasks, integrations, analytics)
- ☐ BPMN 2.0 diagram support
- ☐ Rules engine (DMN-style decision tables)
- ☐ Kafka/RabbitMQ integration
- ☐ Auth (OAuth2/JWT) and multi-tenancy
- ☐ Dashboard & analytics API
- ☐ Visual BPMN designer (drag-and-drop, web-based)

Contributing

Contributions are welcome. Please open an issue to discuss significant changes before submitting a PR.

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